

Optimal Asset Allocation under HARA Utility with Regret Relative to Benchmarked Contributions

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Abstract

This paper studies optimal asset allocation in a continuous-time setting where the investor has hyperbolic absolute risk aversion (HARA) utility and experiences regret relative to a benchmark defined by cumulative net contributions. The model captures behavioral frictions in pension investing and derives the optimal portfolio strategy under regret aversion. Analytical results and comparative statics highlight how regret reshapes risk-taking relative to benchmark proximity.

1 Introduction

Why do some investors remain under-invested in equity markets, even with long horizons? How should optimal asset allocation reflect not only risk preferences but also psychological costs of underperforming a reference point, such as cumulative contributions? This paper tackles these questions by embedding regret relative to a benchmark into a continuous-time portfolio model with HARA utility.

We depart from classical Merton-style models that assume time-separable preferences and rational agents optimizing expected utility over terminal wealth (Merton, 1969; Merton, 1971). While elegant, these models omit behavioral dynamics such as regret aversion, which plays a significant role in real-world investing, especially in defined contribution pension schemes. Behavioral finance has long recognized that investors derive utility not just from outcomes, but from how outcomes compare to reference points (Bell, 1982; Loomes and Sugden, 1982; Barberis, 2013).

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We introduce regret over terminal wealth relative to a benchmark defined by cumulative contributions, capturing the psychological loss investors feel when final wealth underperforms the capital they themselves invested. This form of regret is intuitive and policy-relevant: underperformance relative to one’s own contributions is a common concern in pension planning.

Using HARA preferences provides flexibility in modeling varying risk aversion across wealth levels (Karatzas and Shreve, 1998; Ingersoll, 1987). We embed the regret-adjusted utility into a dynamic optimization framework and solve for the optimal risky asset share using the Hamilton–Jacobi–Bellman (HJB) equation.

1.1 Our Contribution and Research Design

We formulate a continuous-time control problem in which regret is driven by the gap between terminal wealth and a benchmark defined by net contributions. Within this framework, we derive closed-form expressions for the optimal risky-asset share under HARA utility with regret aversion. Our comparative statics show that as an investor’s wealth approaches (or falls below) the benchmark, benchmark proximity effectively amplifies risk aversion, leading to lower equity exposure. We conclude by outlining implications of these results for pension design and lifecycle investing.

2 Literature Review

Classical portfolio theory begins with (Merton, 1969; Merton, 1971), where investors maximize expected utility of terminal wealth. Extensions include stochastic labor income DybvigLiu2010 and habit formation. HARA preferences generalize CRRA/constant relative risk aversion utility and permit analytical tractability (Ingersoll, 1987). Behavioral extensions to portfolio choice include regret aversion (Bell, 1982; Loomes and Sugden, 1982), loss aversion, and probability weighting. Applications include regret over missing market rallies, timing errors, or underperformance relative to benchmarks. (Basak and Makarov, 2014; Gong et al., 2020) develop strategic allocation models incorporating regret or benchmarking, while (Wu and Zeng, 2015) provide continuous-time formulations. Our paper builds on this by modeling regret relative to net contributions—a psychologically and contractually salient benchmark in pension systems.

3 Model Setup

3.1 Financial Market

We work in continuous time with a risk-free asset G_t and a risky asset S_t . The economy at time t is summarized by (R_t, V_t, W_t, M_t) : R_t is the short rate governing the risk-free return, V_t is the risky asset's instantaneous variance, W_t is the investor's wealth, and M_t is a benchmark tracking cumulative contributions (used for performance/regret comparisons). Since the objective depends on wealth relative to the benchmark, define the gap $X_t := W_t - M_t$.

3.2 Risk-free Asset: Short Rate (CIR).

We model the instantaneous short rate R_t using:

$$dR_t = \kappa_r(\theta_r - R_t) dt + \sigma_r \sqrt{R_t} dZ_t^r, \quad (1)$$

where:

- R_t is the instantaneous short rate,
- $\kappa_r > 0$ is the mean-reversion speed of the short rate process,
- $\theta_r > 0$ is the long-run equilibrium interest rate,
- $\sigma_r > 0$ is the volatility of rate shocks, and
- Z_t^r is the standard Brownian motion driving the short-rate shocks.

3.3 Risky Asset: Heston Model (CIR Variance).

Next, we model the risky asset.

3.3.1 Variance

We model the instantaneous variance V_t using:

$$dV_t = \kappa_v(\theta_v - V_t) dt + \sigma_v \sqrt{V_t} dZ_t^v, \quad (2)$$

where:

- V_t is the instantaneous variance of the risky asset,

- $\kappa_v > 0$ is the mean-reversion speed of the variance process,
- $\theta_v > 0$ is the long-run equilibrium level of variance,
- $\sigma_v > 0$ is the volatility of variance shocks, and
- Z_t^v is the standard Brownian motion driving the variance shocks.

3.3.2 Return

We model the risky asset price process S_t under stochastic variance V_t as:

$$\frac{dS_t}{S_t} = \mu_S dt + \sqrt{V_t} dZ_t^S, \quad (3)$$

where:

- S_t is the risky asset price,
- μ_S is the (constant) expected return of the risky asset,
- V_t is the instantaneous variance process, and
- Z_t^S is the standard Brownian motion driving risky-asset return shocks.

3.3.3 Correlations

Brownian motions have constant correlations:

$$d\langle Z^r, Z^v \rangle_t = \rho_{rv} dt, \quad d\langle Z^S, Z^r \rangle_t = \rho_{Sr} dt, \quad d\langle Z^S, Z^v \rangle_t = \rho_{Sv} dt,$$

with $|\rho_{rv}|, |\rho_{Sr}|, |\rho_{Sv}| \leq 1$.

3.3.4 Control

The control is the *dollar amount* in the risky asset, π_t (equivalently, the risky *fraction* $\alpha_t = \pi_t/W_t$).

3.4 Contribution Benchmark

We define the cumulative net contribution benchmark M_t by

$$M_t := M_0 + \int_0^t \left[c_0 + \eta (W_s - M_s)^+ - \xi (M_s - W_s)^+ \right] ds. \quad (4)$$

We use the performance-based rule:

$$dM_t = [c_0 + \eta(W_t - M_t)^+ - \xi(M_t - W_t)^+] dt \quad \text{with} \quad M_0 \geq 0, \quad (5)$$

where:

- M_t is the cumulative net contribution benchmark,
- W_t is the investor's wealth at time t ,
- $c_0 \geq 0$ is the baseline contribution rate,
- $\eta \geq 0$ is the upward sensitivity when $W_t > M_t$ (optimism), and
- $\xi \geq 0$ is the downward sensitivity when $M_t > W_t$ (pessimism).¹

3.5 Wealth Dynamics

The portfolio evolves with an external flow dM_t credited to the account, so the investor's wealth W_t evolves according to:

$$dW_t = [R_t(W_t - \pi_t) + \mu_S \pi_t] dt + \sqrt{V_t} \pi_t dZ_t^S + dM_t, \quad (6)$$

where:

- π_t is the dollar amount invested in the risky asset at time t ,
- R_t is the instantaneous short rate at time t ,
- μ_S is the (constant) expected return of the risky asset,
- V_t is the instantaneous variance of the risky asset at time t , and
- Z_t^S is the standard Brownian motion driving risky-asset return shocks.

Making the substitution $\alpha_t = \pi_t/W_t$ yields:

$$dW_t = [R_t(1 - \alpha_t)W_t + \mu_S \alpha_t W_t] dt + \sqrt{V_t} \alpha_t W_t dZ_t^S + dM_t, \quad (7)$$

where:

- $r_t(1 - \alpha_t)W_t$ represents the safe asset return,

¹ $(x)^+ = \max(x, 0)$

- $\mu_S \alpha_t W_t$ represents the risky asset return,
- dM_t is the net external cash flow into the account (a contribution when $dM_t > 0$ and a withdrawal when $dM_t < 0$), and
- $\sqrt{v_t} \alpha_t W_t dZ_t^S$ represents the random shock from equity risk.

3.6 Utility Function with Regret

We adopt a HARA utility function:

$$U(W_\tau - M_\tau) = \frac{1}{1-\gamma} (a(W_\tau - M_\tau) + b)^{1-\gamma} \text{ with } a > 0, b \geq 0.$$

After simplification, we have:

$$U(X) = \frac{(aX + b)^{1-\gamma}}{1-\gamma} \text{ with } \{X : aX + b > 0\} \text{ and } \gamma \neq 1, \quad (8)$$

where:

- γ is the risk-aversion parameter (HARA form) and
- a, b are the scaling and shift constants.²

4 Problem Formulation and HJB Equation

We study a continuous-time portfolio choice problem over a finite horizon $[0, T]$ in a stochastic investment opportunity set with CIR short rate and Heston stochastic variance. The investor controls the risky position through an admissible, progressively measurable strategy π_t . The investor's wealth W_t evolves endogenously through trading returns and an external net cash-flow process. The benchmark M_t is defined as the cumulative net contribution/withdrawal ledger and evolves as a finite-variation process whose drift depends on the current performance gap $W_t - M_t$. As a result, the controlled state vector is

$$(R_t, V_t, M_t, W_t)$$

and the system is Markov in this augmented state.

²The benchmark M_τ induces regret aversion when W_τ is near M_τ .

The investor evaluates terminal performance relative to the contribution benchmark via HARA utility defined on the gap $W_t - M_t$. Given an initial condition (t, r, v, w, m) . We define the value function:

$$J(t, r, v, w, m) = \sup_{\pi} \mathbb{E}_{t,r,v,w,m} [U(W_T - M_T)]. \quad (9)$$

With terminal condition: $J(T, r, v, w, m) = U(w - m)$.

Assumption 1 (Well-posedness baseline). $\kappa_r, \theta_r, \sigma_r, \kappa_v, \theta_v, \sigma_v > 0$ satisfy Feller conditions $2\kappa_r\theta_r \geq \sigma_r^2$ and $2\kappa_v\theta_v \geq \sigma_v^2$; correlations are constant; μ_S is constant; $c_0, \eta, \xi \geq 0$. Initial state (R_0, V_0, W_0, M_0) lies in the natural domain with $a(W_0 - M_0) + b > 0$.

Definition 1 (Admissible control). A progressively measurable π is admissible if it satisfies either a fraction constraint $\alpha_t = \pi_t/W_t \in [0, 1]$ (with $\pi_t \equiv 0$ when $W_t = 0$) or a linear-growth bound $|\pi_t| \leq K(1 + W_t)$ a.s. for some $K > 0$, and the state SDEs admit a unique strong solution with

$$\mathbb{E} \left[\sup_{s \leq T} (|W_s|^p + M_s^p + R_s^p + V_s^p) \right] < \infty \quad \text{for some } p > 1.$$

By the dynamic programming principle, J satisfies a Hamilton–Jacobi–Bellman (HJB) equation obtained by applying Itô’s formula to $J(t, R_t, V_t, M_t, W_t)$.

Lemma 1 (Itô for $J(t, R, V, W, M)$). If $J \in C^{1,2,2,2,2}$, then

$$\begin{aligned} dJ &= J_t dt + J_r dR + J_v dV + J_w dW + J_m dM \\ &\quad + \frac{1}{2} J_{rr} (dR)^2 + \frac{1}{2} J_{vv} (dV)^2 + \frac{1}{2} J_{ww} (dW)^2 \\ &\quad + J_{rv} dR dV + J_{rw} dR dW + J_{vw} dV dW. \end{aligned}$$

With (1)–(7), we have the quadratic covariations

$$\begin{aligned} (dR)^2 &= \sigma_r^2 R dt, & (dV)^2 &= \sigma_v^2 V dt, & (dW)^2 &= V \pi^2 dt, \\ dR dV &= \rho_{rv} \sigma_r \sigma_v \sqrt{RV} dt, & dW dR &= \rho_{Sr} \sigma_r \sqrt{RV} \pi dt, & dW dV &= \rho_{Sv} \sigma_v V \pi dt. \end{aligned} \quad (10)$$

Substitute dR, dV, dW, dM into dJ and collect dt terms. Write the drifts and diffusions explicitly:

$$\begin{aligned} dR &= \underbrace{\kappa_r(\theta_r - R)}_{\mu_R} dt + \underbrace{\sigma_r \sqrt{R}}_{\sigma_R} dZ^r, & dV &= \underbrace{\kappa_v(\theta_v - V)}_{\mu_V} dt + \underbrace{\sigma_v \sqrt{V}}_{\sigma_V} dZ^v, \\ dW &= \underbrace{R(W - \pi) + \mu_S \pi}_{\mu_W} dt + \underbrace{\sqrt{V} \pi}_{\sigma_W} dW^S + \underbrace{\dot{M}}_{\mu_M} dt, & dM &= \dot{M} dt. \end{aligned}$$

Thus the *drift* (the dt -coefficient) of dJ is

$$\begin{aligned} \text{Drift}(dJ) = & J_t + \mu_R J_r + \frac{1}{2} \sigma_R^2 J_{rr} + \mu_V J_v + \frac{1}{2} \sigma_V^2 J_{vv} + \rho_{rv} \sigma_R \sigma_V J_{rv} \\ & + \mu_W J_w + \frac{1}{2} \sigma_W^2 J_{ww} + \rho_{Sr} \sigma_R \sigma_W J_{rw} + \rho_{Sv} \sigma_V \sigma_W J_{vw} \\ & + \underbrace{\dot{M} J_w + \dot{M} J_m}_{\text{from } dW \text{ (inflow) and } dM}. \end{aligned}$$

Plug back $\mu_R, \sigma_R, \mu_V, \sigma_V, \mu_W, \sigma_W$ to obtain:

$$\begin{aligned} \text{Drift}(dJ) = & J_t + \kappa_r(\theta_r - r) J_r + \frac{1}{2} \sigma_r^2 J_{rr} + \kappa_v(\theta_v - v) J_v + \frac{1}{2} \sigma_v^2 J_{vv} + \rho_{rv} \sigma_r \sigma_v \sqrt{rv} J_{rv} \\ & + [r(w - \pi) + \mu_S \pi] J_w + \frac{1}{2} v \pi^2 J_{ww} + \rho_{Sr} \sigma_r \sqrt{rv} \pi J_{rw} + \rho_{Sv} \sigma_v v \pi J_{vw} \\ & + \dot{M} J_w + \dot{M} J_m. \end{aligned}$$

Hamilton–Jacobi–Bellman Equation: Under Assumptions 1 and admissible π , By the Bellman principle, the drift of J under the optimal control must vanish:

$$0 = \sup_{\pi} \text{Drift}(dJ),$$

Which gives the HJB:

$$\begin{aligned} 0 = \sup_{\pi} \Big\{ & J_t + \kappa_r(\theta_r - r) J_r + \frac{1}{2} \sigma_r^2 J_{rr} + \kappa_v(\theta_v - v) J_v + \frac{1}{2} \sigma_v^2 J_{vv} + \rho_{rv} \sigma_r \sigma_v \sqrt{rv} J_{rv} \\ & + [r(w - \pi) + \mu_S \pi] J_w + \frac{1}{2} v \pi^2 J_{ww} + \rho_{Sr} \sigma_r \sqrt{rv} \pi J_{rw} + \rho_{Sv} \sigma_v v \pi J_{vw} \\ & + \dot{M} J_w + \dot{M} J_m \Big\}, \quad J(T, r, v, w, m) = U(w - m). \end{aligned} \quad (11)$$

, with $\dot{M} := c_0 + \eta(w - m)^+ - \xi(m - w)^+$ and terminal condition $J(T, r, v, w, m) = U(w - m)$.

Remark 1: Where the benchmark cancels in the driver. If preferences depend on the *gap* $X = w - m$ only, write $J(t, r, v, w, m) = \tilde{J}(t, r, v, X)$. Then $J_w = \tilde{J}_X$ and $J_m = -\tilde{J}_X$, hence $\dot{M}(J_w + J_m) \equiv 0$. The benchmark still affects the *state* (hence future risk tolerance), but not the instantaneous HJB driver.

Concavity in π . Since $v > 0$ and $J_{ww} < 0$ under risk aversion, the Hamiltonian is strictly concave in π and the supremum is attained by the FOC.

5 Solution and Optimal Strategy

Group the π -dependent terms in the Hamiltonian:

$$\mathcal{H}(\pi) = \underbrace{(\mu_S - r)J_w}_{\text{Sharpe channel}} \pi + \frac{1}{2}v J_{ww} \pi^2 + \underbrace{\rho_{Sr}\sigma_r\sqrt{rv} J_{rw}}_{\text{term-structure hedge}} \pi + \underbrace{\rho_{Sv}\sigma_v v J_{vw}}_{\text{volatility hedge}} \pi + (\text{terms independent of } \pi).$$

FOC: $\partial\mathcal{H}/\partial\pi = 0$ gives

$$(\mu_S - r)J_w + v \pi J_{ww} + \rho_{Sr}\sigma_r\sqrt{rv} J_{rw} + \rho_{Sv}\sigma_v v J_{vw} = 0,$$

so

$$\pi^* = - \frac{(\mu_S - r)J_w + \rho_{Sr}\sigma_r\sqrt{rv} J_{rw} + \rho_{Sv}\sigma_v v J_{vw}}{v J_{ww}}. \quad (12)$$

If portfolio constraints apply (e.g. $\alpha = \pi/W \in [0, 1]$), the optimal control is the Euclidean projection of (12) onto the feasible set.

5.1 Homothetic Reduction (HARA-in-the-Gap)

We posit the homothetic form

$$J(t, r, v, w, m) = \frac{[a(w - m) + b]^{1-\gamma}}{1-\gamma} G(t, r, v), \quad G(T, r, v) = 1, \quad G > 0. \quad (13)$$

Let $X := w - m$. Then

$$\begin{aligned} J_w &= a[aX + b]^{-\gamma} G, & J_{ww} &= -\gamma a^2 [aX + b]^{-\gamma-1} G, \\ J_{rw} &= a[aX + b]^{-\gamma} G_r, & J_{vw} &= a[aX + b]^{-\gamma} G_v, & J_m &= -J_w. \end{aligned} \quad (14)$$

Definition 2 (Risk tolerance in the gap). The (dollar) risk tolerance associated with the gap X is

$$\mathcal{R}(X) := -\frac{J_w}{J_{ww}} = \frac{aX + b}{\gamma a}.$$

Proposition 1 (Closed-form feedback control). *Substituting (14) into (12) and simplifying gives*

$$\pi^*(t, r, v, X) = \mathcal{R}(X) \left[\frac{\mu_S - r}{v} + \rho_{Sr}\sigma_r\sqrt{\frac{r}{v}} \frac{G_r}{G} + \rho_{Sv}\sigma_v \frac{G_v}{G} \right]. \quad (15)$$

The risky fraction is $\alpha^* = \pi^*/W$ (project to $[0, 1]$ if constrained).

Remark 2: Benchmark proximity amplifies risk aversion. As $X \downarrow 0$ (wealth approaches benchmark), $\mathcal{R}(X) \downarrow$ and the dollar risky position shrinks mechanically. This is the direct, instantaneous regret channel. Intertemporally, M_t also shifts the drift of X_t , altering future $\mathcal{R}(X)$ and thus future allocations.

5.2 Reduced PDE for the Multiplier G

TODO: Connecting Sentences

6 Comparative Statics and Insights

7 Numerical Illustration

Simulate sample paths, plot optimal α_t^* as a function of $W_t - M_\tau$, perform sensitivity to γ , M_τ , and a .

8 Conclusion

We show that regret relative to contributions distorts optimal asset allocation,

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